



Bincom Dev Center

BINCOM ACADEMY CURRICULUM

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CYBERSECURITY CLASS CURRICULUM

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DAY 1: Network Security Fundamentals & Lab Setup

Lesson: Threat Landscape and Attack Surface

- Understanding the evolving cybersecurity landscape
- Identifying attack surfaces and entry points
- Types of network attacks: DDoS, Man-in-the-Middle (MITM), sniffing, and port scanning

Lesson: Defense Layers and Core Tools

- Firewalls, IDS, IPS, and Honeypots
- Overview of cybersecurity tools: Kali Linux, Nmap, Wireshark, Cowrie Honeypot, HoneyDB

Lesson: Lab Setup

- Setting up a local cybersecurity lab using VirtualBox or VMware

- Installing Kali Linux and other security tools

Lesson: Cybersecurity Essentials

- CIA Triad (Confidentiality, Integrity, Availability)
- AAA (Authentication, Authorization, Accounting)
- Principle of Least Privilege and Defense in Depth
- Basic cryptography concepts (Symmetric vs Asymmetric, Hashing, TLS/SSL)
- Common security controls: MFA, patching, backups
- Secure coding fundamentals

Links to Online Study:

- <https://www.youtube.com/watch?v=zIWGjkr0ENE>
- Cowrie Honeypot GitHub Repo
- https://www.youtube.com/watch?v=1O9y8Rfyc6s&list=PL_o0Qzu-8MKhg6co37kssx6lRdQ4o1YpX

Assignment 1: *Network Attack Detection & Reporting*

- Install Kali Linux and Cowrie Honeypot on separate VMs
- Use Nmap to perform scanning and basic attacks
- Capture network traffic with Wireshark
- Prepare a 1-page incident report

Deliverables:

- Screenshot of honeypot running
- Nmap scan results

- .pcap file
- 1-page incident report

DAY 2: Vulnerability Assessment and Exploitation

Lesson: Vulnerabilities and Exploitation

- Understanding CVE, CVSS scoring
- Vulnerability scanning workflow
- Tools overview: Nessus, OpenVAS, Nikto, Metasploit

Lesson: Exploitation in Controlled Labs

- https://www.youtube.com/watch?v=eyeuKa6v_B8
- <https://www.youtube.com/watch?v=iLdsCnpMnTg>
- Exploiting common web vulnerabilities

Links to Online Study:

- Metasploitable2 Download
- OpenVAS Setup Guide
- DVWA Setup Instructions

Assignment 2: *Vulnerability Scan and Exploit Simulation*

- Set up DVWA
- Run a vulnerability scan using OpenVAS
- Exploit one vulnerability using Metasploit
- Write a remediation plan

Deliverables:

- Vulnerability scan report
- Screenshot of successful exploit
- Mitigation/Remediation document

DAY 3: Web Application and Endpoint Security

Lesson: Web Application Security

- OWASP Top 10 vulnerabilities (Injection, XSS, CSRF, etc.)
- Using Burp Suite and OWASP ZAP for web testing

Lesson: Endpoint & Application Defense

- Endpoint security: EDR, patch management
- Web Application Firewall (WAF) using ModSecurity
- ISO/IEC 27001 (Security management systems)

Lesson: Application Hardening (PrestaShop Case Study)

- Apply security checklist for PrestaShop
- Simulate attacks and verify defenses

Links to Online Study:

- OWASP ZAP Lab – TryHackMe
- Burp Suite Community Edition
- ModSecurity Configuration Guide

Assignment 3: *Attack and Defend Web Applications*

- <https://www.youtube.com/watch?v=cwUrc7JL1hE&list=PLMcXv2jVcbgp4J7240jF3pxGh8LIsHdCU>

- <https://www.youtube.com/watch?v=MB7nQrIP5Yc>
- Deploy or use PrestaShop for security assessment

Deliverables:

- Attack and defense screenshots
- WAF rule file
- Report on security validation
- Completed PrestaShop security checklist and report

DAY 4: Security Monitoring & Incident Response

Lesson: Security Monitoring Concepts

- <https://www.youtube.com/watch?v=3JTjVa0IM7o&list=PLCthQV7PRfBZORsXWFe-B2JCQpzftnvDb>
- <https://www.youtube.com/watch?v=6o78i9QFcF4&list=PLyJqGMYm0vnMHLPbmT1-fknzTzR7auBEb>

Lesson: Incident Response Frameworks

- NIST & MITRE ATT&CK
- Risk Management Frameworks (NIST RMF, ISO 31000)

Lesson: AI Risk in Cybersecurity

- Securing AI agents and ML models
- AI risk assessment: data poisoning, prompt injection, misuse detection
- Integrating AI risk into incident response

Links to Online Study:

- Security Onion Documentation
- MITRE ATT&CK Framework
- Splunk Free Training

Assignment 4: *Log Analysis & Attack Detection*

- Deploy Security Onion or ELK Stack
- Simulate brute-force SSH attack using Hydra
- Detect and analyze using SIEM
- Perform AI risk assessment on an adversarial ML tool

Deliverables:

- SIEM dashboard screenshot
- Incident analysis report
- AI tool justification and risk assessment report

DAY 5: Cloud Security & Identity Access Management (IAM)

Lesson: Cloud Security Essentials

- Common cloud threats: misconfiguration, privilege escalation
- IAM best practices: MFA, RBAC, least privilege

Lesson: Practical Cloud Defense

- Simulating IAM misconfiguration and privilege escalation
- Applying secure IAM policies

Links to Online Study:

- AWS IAM Documentation(<https://docs.aws.amazon.com/iam/>)
- <https://www.youtube.com/watch?v=wTUyH6mk2Ms>
- CloudGoat – AWS PenTesting Toolkit

Assignment 5: Cloud Privilege Escalation Simulation

- Create AWS IAM user with misconfigured permissions
- Perform privilege escalation
- Fix with least privilege configuration

Deliverables:

- IAM screenshots
- CLI logs
- Documentation of vulnerability and fix

DAY 6: Revision & Capture the Flag (CTF)

Lesson: Review and Practical Application

- Recap of all previous sessions
- Introduction to Capture the Flag (CTF) environments (TryHackMe, HackTheBox)

Activity: Team-based CTF Challenge

- Tasks: Reconnaissance, exploitation, privilege escalation, reporting

Links to Online Study:

- <https://www.youtube.com/watch?v=VcBiJuu-8LU>

- <https://www.youtube.com/watch?v=3r6RiyxVccU&list=PLvrX-Ob08KZJ6rAVcauUPBVNS6puaYtI9>

Assignment 6: *Capture the Flag (CTF)*

- Capture at least 3 flags in a designated CTF environment
- Document findings and methodology

Deliverables:

- CTF flags
- CTF writeup document

DAY 7: Test

Comprehensive test covering:

- Network Security
- Vulnerability Management
- Web Application Security
- Incident Response
- Cloud Security

DAY 8: Review & Presentation

Lesson: Review & Project Presentation

- Review of test results